

Stanley College of Engineering & Technology for Women (A)**B.E (ECE) IV-Semester (Main) Examinations-July-2023****Integrated Circuits and Applications**

Time: 03Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. Explain term Non linear wave shaping and which circuits will come under this category. [2M CO1 L4]
- b. Illustrate Clamping circuit theorem with the help of expression. [2M CO1 L3]
- c. Analyze how the commutating capacitors will increase the speed of a fixed-bias binary. [2M CO2 L4]
- d. Define differential amplifier [2M CO3 L1]
- e. Op-amp is popular. Comment [2M CO4 L5]
- f. Draw the functional diagram of IC-555 [2M CO5 L4]

Part-B**4X12=48M**

2. a) Explain the effect of diode characteristics on clamping voltage. [6M CO1 L2]
b) List out the applications of Comparators. [6M CO1 L1]
3. a) Derive Rise time of a Low Pass RC Circuit. [6M CO1 L4]
b) Sketch the basic circuit diagram of a two level clipper circuit & explain its operation. [6M CO1 L4]
4. a) Analyze the circuit of Bistable Multivibrator with necessary waveforms. [6M CO2 L4]
b) Show how Astable works as Voltage to Time converter with the help of necessary derivation and neat diagram. [6M CO2 L3]
5. a) Explain the DC characteristics of OPAMP. [6M CO3 L2]
b) What are different configurations of differential amplifier? Draw the neat-labeled diagram for a) dual input, balanced output, b) dual input, unbalanced output and c) single input, balanced output [6M CO3 L5]
6. a) Design first order High Pass filter using OPAMP. [6M CO4 L2]
b) Draw the circuit symbol of op-amp. Explain what is mean by inverting input and non-inverting input. [6M CO4 L2]
7. a) Explain Monostable Multivibrator using IC 555. [6M CO5 L3]
b) Write short notes on Flash-type A/D converter. [6M CO5 L3]

Hall Ticket No. -

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Stanley College of Engineering & Technology for Women (A)

B.E (ECE) IV Semester (Main) Examinations-July-2023

Computer Organization and Architecture

Time: 03Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. Distinguish computer organization and architecture. [2M CO1 BTL2]
- b. Define FETCH and Decode operations. [2M CO2 BTL1]
- c. Draw the flowchart of Source Initiated Handshaking Transfer Communication. [2M CO3 BTL3]
- d. Distinguish Programmed I/O, Interrupt driven I/O. [2M CO4 BTL2]
- e. Draw D-RAM memory cell. [2M CO5 BTL3]
- f. MOV [2050],AX interpret the addressing mode of this instruction. [2M CO2 BTL3]

Part-B

4X12=48M

2. a) Explain the Bus architecture of Von-Neumann. [6M CO1 BTL3]
- b) Discuss about Multiplication using Booth's algorithm with an example. [6M CO1 BTL3]
3. a) Describe Micro programmed Control Organization. [6M CO2 BTL3]
- b) Explain about the instructions with memory operand with examples. [6M CO2 BTL3]
4. a) Distinguish CISC and RISC processors in detail. [6M CO3 BTL2]
- b) Explain about the Instruction Pipeline with an example instruction. [6M CO3 BTL3]
5. a) Describe the differences between Programmed I/O, Interrupt driven I/O. [6M CO4 BTL3]
- b) Explain about DMA master and slave modes. [6M CO4 BTL3]
6. a) Describe memory hierarchy and explain its advantages. [6M CO5 BTL3]
- b) Explain about paging concept with an example. [6M CO5 BTL3]
7. a) List out the advantages of pipelining mechanism. [4M CO3 BTL1]
- b) Explain about Pipeline and Vector Processing. [8M CO3 BTL3]

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Stanley College of Engineering & Technology for Women (A)

B.E (ECE) IV Semester (Main) Examinations-July-2023

Signals and Systems

Time: 03Hours

Max. Marks-60

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Part-A

6X2=12M

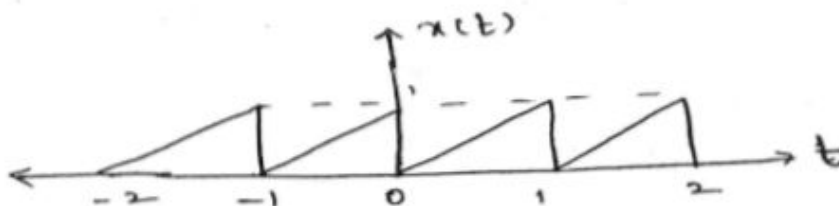
1. a. Define Unit impulse signal and write its properties. [2M CO1 BTL1]
 b. Sketch the following signal [2M CO1 BTL3]
 (i) $2u(t+2)-2u(t-3)$ (ii) $r(t)+r(t+2)-r(t+1)$
 c. State Dirichlet's conditions for Fourier series of a continuous time periodic signal. [2M CO2 BTL1]
 d. Find the Fourier transform of (i) $\delta(n)$ (ii) $\delta(n-k)$? [2M CO2]
 e. Define Initial value and Final value theorem in Laplace transform. [2M]
 f. State sampling theorem. [2M]

Part-B

4X12=48M

2. a) Explain the basic operations that can be performed on continuous-time signal with an example. [6M CO1 BTL2]
 b) Comment about linearity, Causality and time invariance of the system [6M CO1 BTL5]

$$y(t) = 2x(t+1) - x(t-1)^2$$
3. a) Obtain the trigonometric Fourier Series for the waveform shown in figure below. [7M CO2 BTL3]



- b) Show that the functions $\sin(n\omega_0 t)$ and $\cos(n\omega_0 t)$ are orthogonal over any interval $\{t_0, t_0 + (2\pi/\omega_0)\}$ [5M CO2 BTL3]
4. a) Find the Fourier transform of the following signals using properties. [6M CO3 BTL3]
 (i) $x(t) = e^{-at}u(t)$
 (ii) $x(t) = \delta(t+2) + \delta(t+1) + \delta(t-1)$
 (iii) $x(t) = e^{-2t}[u(t) - u(t-5)] + \delta(t-2)$

b) Find DTFT of the following

[6M CO3 BTL3]

(i) $x(n) = \{1, -2, 2, 3\}$ (ii) $x(n) = (1/2)^{n-2} u(n-2)$

5. a) The input and output of a Causal LTI system is described by

[6M CO4 BTL3]

$$d^2y/dt^2 + 5dy/dt + 6y(t) = x(t) . \text{ Find the impulse response of the system}$$

b) Find the inverse Laplace transform of

[6M CO4 BTL3]

$$X(S) = S/(S^2 + a^2)^2$$

6.a) Find the Z-transform of the following sequences

[6M CO5 BTL3]

(i) $x(n) = u(n) - u(n-4)$ (ii) $x(n) = \{-1, -2, -3, 1, 2, 3\}$

b) Determine the Convolution Sum of the following sequence

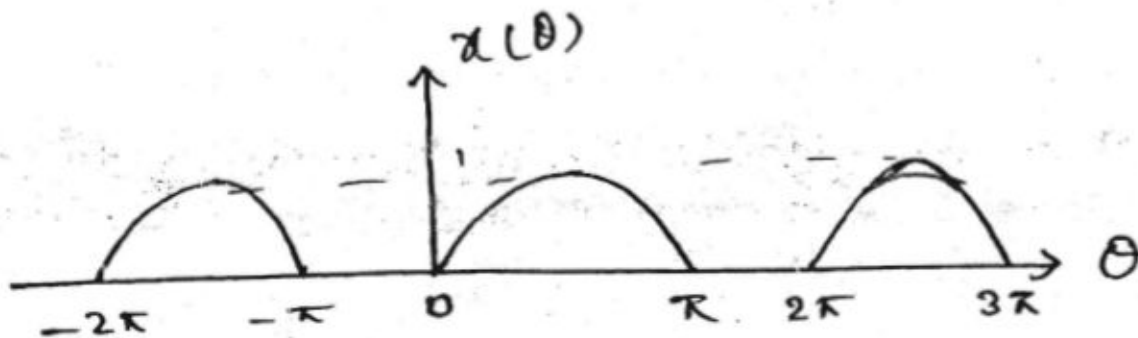
[6M CO5 BTL3]

(i) $x(n) = \{1, 2, -1, 3\}$ (ii) $x(n) = \{0, 1, 2, -1\}$

$h(n) = \{1, 2, 0, -2\}$ $h(n) = \{-1, 2, 1, 3\}$

7.a) Obtain the exponential Fourier series for the waveform shown in figure below

[8M CO2 BTL3]



b) Find the Fourier transform of the signum function

[4M CO3 BTL3]

Stanley College of Engineering & Technology for Women (A)**B.E (ECE) IV Semester (Main) Examinations-July-2023****Antennas and Wave Propagation****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining SIX questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. Define the terms as applicable to antennas
 (a) Beam width and (b) Beam area [2M, CO1, BTL1]
 b. Differentiate between half-wave dipole and quarter wave monopole [2M, CO2, BTL2]
 c. Discuss the role of Reflector, Dipole and Director in Yagi-Uda antenna [2M, CO3, BTL1]
 d. Discuss the types of horn antennas [2M, CO3, BTL2]
 e. Discuss few Radiation Hazards [2M, CO4, BTL2]
 f. Describe Line of sight propagation [2M, CO5, BTL1]

Part-B**4X12=48M**

2. a. Explain the effective length and effective area of the antenna. [6M, CO1, BTL2]
 b. For an antenna radiating with radiation intensities of (i) $2\cos\theta$ [6M, CO1, BTL2]
 (ii) $2\cos3\theta$. Estimate the directivities of the antenna.
3. a. Analyse mathematically the far field pattern of circular loop with [6M, CO2, BTL3]
 uniform current
 b. Describe the practical design considerations for monofilar helical antenna [6M, CO2, BTL4]
4. a. Describe the construction of a pyramidal horn and draw its radiation [6M, CO3, BTL3]
 pattern
 b. Discuss the properties of parabolic reflector antennas with centre fed [6M, CO3, BTL4]
 and cassegrain feed types and show that the gain of the same is $G = 6:0 (D/\lambda)^2$
5. a. Derive an expression for the radiation field of an 'n' element linear [6M, CO4, BTL3]
 array with spacing 'd'.

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- b. Compare the performance & properties of 'n' element linear arrays and binomial arrays of 'n' elements. [6M, CO4, BTL4]
6. a. Determine the field due to array of two point sources of same amplitude and opposite phase at a distance 'R'. [6M, CO4, BTL3]
- b. Describe the neat set up for measuring the gain of an antenna. [6M, CO4, BTL4]
7. a. A radio link has a 15W transmitter connected to an antenna of 2.5m^2 effective aperture at 5GHz. The receiving antenna has an effective aperture of 0.5 m^2 located at a 15Km LOS distance from the transmitting antenna. Assuming lossless, matched antennas, find the power delivered to the receiver [6M, CO5, BTL2]
- b. Explain refraction and reflection of sky waves by ionosphere [6M]



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Stanley College of Engineering & Technology for Women (A)

B.E (ECE) IV Semester (Main) Examinations-July-2023

Analog Electronic Circuits

Time: 03 Hours

Max. Marks-60

- Note:**
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Part-A

6X2=12M

- Interpret major difference between a BJT and JFET amplifiers? [2M CO1 BTL1]
 - Define the parameters. Gain, Bandwidth, i/p & o/p impedance of ideal amplifier [2M CO2 BTL2]
 - Explain the concept of Barkhausen criteria [2M CO3 BTL2]
 - Differentiate between Class-A, B, AB, C & D amplifiers [2M CO4 BTL2]
 - Explain the harmonic distortion in power amplifiers [2M CO4 BTL1]
 - Write the relation between gain and bandwidth product [2M CO1 BTL1]

Part-B

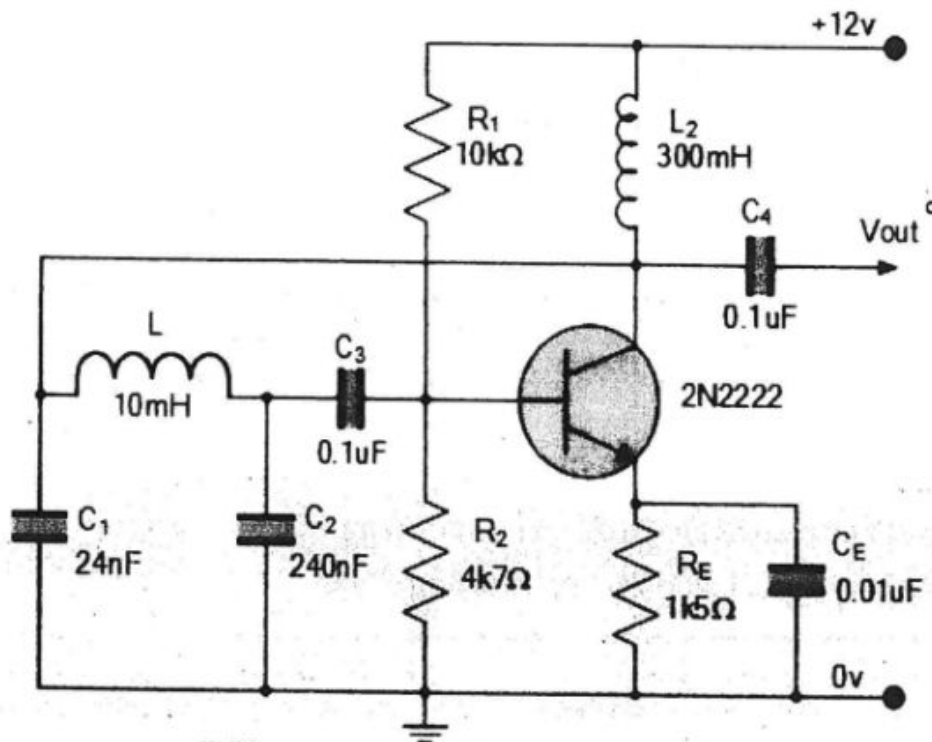
4X12=48M

- If an amplifier with gain of - 1000 and feedback factor $\beta = - 0.1$ had a gain change of 20% due to temperature, determine the change in gain of the feedback amplifier [6M CO2 BTL3]
 - Describe the analysis of single stage RC coupled amplifier [6M CO1 BTL4]
- Discuss the effect of negative feedback on input and output impedances [6M CO3 BTL1]
 - Discuss the general characteristics of negative feedback
- Analyze how negative and positive feedback is enhanced in amplifiers with a neat diagram [6M CO3 BTL4]
 - Describe the working model of Wein-bridge oscillator. And mathematically formulate the condition for oscillation. [6M CO3 BTL3]
- The Hartley Oscillator circuit having two individual inductors of 0.5mH each is designed to resonate in parallel with a variable capacitor that can be adjusted between 100pF and 500pF. Determine the upper and lower frequencies of oscillation and also the Hartley oscillator's bandwidth. [6M CO3 BTL3]

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- b. A Colpitts Oscillator having two capacitors of 24nF and 240nF respectively are connected in parallel with an inductor of 10mH . Determine the frequency of oscillations of the circuit [6M CO3 BTL3]



- 6 a. Describe the working model of transformer coupled class A amplifier and find its efficiency [6M CO4 BTL3]
- b. Describe the working model of class B push pull amplifier and calculate its efficiency [6M CO4 BTL3]
- 7 a. Design a single tuned amplifier circuit for 5KHz [6M CO5 BTL4]
- b. Describe how selectivity, gain and bandwidth can be improved in single tuned and double tuned amplifiers [6M CO5 BTL3]

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Code: R123745

Stanley College of Engineering & Technology for Women (A)**B.E (ECE) IV Semester (Main) Examinations-July-2023****Signals and Systems****Time: 03Hours****Max. Marks-60**

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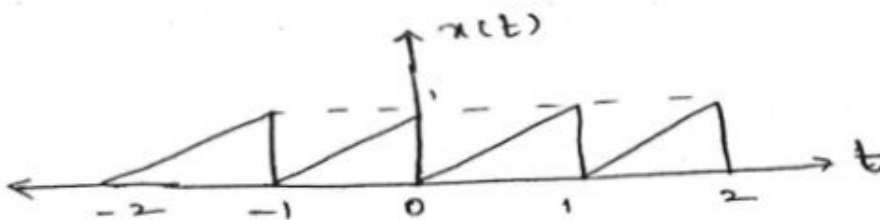
Part-A**6X2=12M**

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- c. State Dirichlet's conditions for Fourier series of a continuous time periodic signal. [2M CO2 BTL1]
- d. Find the Fourier transform of (i) $\delta(n)$ (ii) $\delta(n-k)$? [2M CO3 BTL3]
- e. Define Initial value and Final value theorem in Laplace transform. [2M CO4 BTL1]
- f. State sampling theorem. [2M CO5 BTL1]

Part-B**4X12=48M**

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- b) Comment about linearity, Causality and time invariance of the system [6M CO1 BTL5]

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- b) Show that the functions $\sin(n\omega_0 t)$ and $\cos(n\omega_0 t)$ are orthogonal over any interval $\{t_0, t_0+(2\pi/\omega_0)\}$ [5M CO2 BTL3]
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 - (ii) $x(t) = \delta(t+2)+\delta(t+1)+\delta(t-1)$
 - (iii) $x(t) = e^{-2t}[u(t)-u(t-5)]+\delta(t-2)$